Status	Finished
Started	Tuesday, 20 October 2026, 4:11 PM
Completed	Tuesday, 20 October 2026, 4:17 PM
Duration	6 mins 31 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Selection sort

b. Heap sort

c. Quick sort

d. Merge sort

The correct answer is:
Merge sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. sort()

b. arrange()

c. sorted()

d. order()

The correct answer is: sort()

Question **3**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sorted()
- b. order()
- c. sort()
- d. arrange()

The correct answer is: sorted()

Question **4**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Rearranging
- b. Sorting
- c. Searching
- d. Merging

The correct answer is: Sorting

Question **5**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Deleting data from a list
- b. Searching for data in a list
- c. Inserting data into a list
- d. Arranging data in a particular format

The correct answer is: Arranging data in a particular format

Question **6**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Decreases the efficiency of algorithms
b. Makes code execution slower
c. Makes data harder to manage
d. Provides a basis for other algorithms to work efficiently

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **7**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Complexity
b. Merging
c. Sorting
d. Searching

The correct answer is:
Complexity

Question **8**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By using the bubble-up method
b. By breaking the input into smaller parts and merging them
c. By comparing elements sequentially
d. By sorting data in a single pass

The correct answer is: By breaking the input into smaller parts and merging them

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It divides the list into two halves, sorts each half, and then merges them
- b. It builds a sorted array one element at a time
- c. It repeatedly finds the minimum element and moves it to the sorted part of the list
- d. It compares adjacent elements and swaps them if necessary

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **10**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Merge Sort
- b. Quick Sort
- c. Bubble Sort
- d. Heap Sort

The correct answer is: Bubble Sort

Question **11**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It does not require recursion
- b. It is less efficient than bubble sort
- c. It sorts data using a single pass
- d. It is based on the divide-and-conquer approach

The correct answer is: It is based on the divide-and-conquer approach

Question **12**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Sorting the entire list sequentially
b. Dividing the input into smaller subproblems
c. Comparing each element with the others
d. Combining sorted sublists

The correct answer is: Dividing the input into smaller subproblems

Question **13**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Extraction
b. Distribution
c. Selection
d. Insertion

The correct answer is:
Insertion

Question **14**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It has no effect on the analysis process
b. It makes the analysis slower
c. It complicates the analysis process
d. It allows for quicker identification of duplicates

The correct answer is: It allows for quicker identification of duplicates

Question **15**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys

b. A sorting algorithm is stable if it preserves the order of all keys

c. A sorting algorithm is stable if it preserves the order of duplicate keys

d. A sorting algorithm is stable if it preserves the order of non-duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys